

Supplementary Material: Controlling Embedding Spaces with Text-Conditioned Transformations

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Supplementary Overview

Section A: Dataset details

Section B: Implementation details

Section C: Additional experiment details

A Dataset Details

ShotBench [9]. ShotBench provides annotations for perceptual, camera-related attributes such as camera angle, shot size, composition, and lighting. Although the dataset was originally introduced for visual question answering, we convert the answer strings into categorical labels for each attribute. A small number of answers contain multiple labels for the same category (e.g., “low angle & dutch angle”); these ambiguous cases are removed to ensure a single ground-truth label per category. We use the official split of 31k training images and 2.7k test images. **Clevr-4 [14].** Clevr-4 is a synthetic benchmark constructed to isolate four independent visual attributes: color, count, shape, and texture. Each rendered image depicts multiple 3D objects, and each attribute category contains 10 uniformly sampled subclasses, giving the dataset a clean, balanced structure ideal for learning controlled transformations. We use the official split of 8.4k training images and 2.1k test images.

Stanford 40 Actions [17]. Stanford40 consists of 9,532 images spanning 40 human actions. Following prior multi-attribute clustering works [7], we also use the additional annotations provided for two auxiliary categories: location (10 classes) and mood (4 classes), each defined for a disjoint subset of 1,000 images. To supply supervision for all three categories during training, we annotate a small set of 532 images from the original Stanford40 training split with labels consistent with the released location and mood definitions. These annotations do not overlap with the evaluation subsets. For evaluation, we follow the same protocol as prior work: action is evaluated on the remaining 9,000 images, location and mood are evaluated on their respective 1,000 image annotated subsets.

FruitSALAD [11]. FruitSALAD is a style-aligned artwork dataset designed to disentangle object identity from artistic appearance. It contains 10 fruit categories and 10 artistic styles, with each fruit-style pairing instantiated by 100 images (for a total of 10,000 samples). This structure makes it well suited for evaluating attribute-aware retrieval. We train with even splits, taking 50 of each pairing instance as training images and the remaining as test images.

FER-2013 [3]. FER-2013 is a facial expression recognition dataset containing 35,887 grayscale face images annotated with 7 emotion classes (anger, disgust, fear, happiness, sadness, surprise, neutral). We use the standard split: 28,709 training images and 3,589 public test images.

Card [18]. Card is a playing-card recognition dataset with 8,029 images, annotated along two independent categories: suit (clubs, spades, diamonds, hearts) and rank (2–10, jack, queen, king, ace). We follow the official split of 7,624 training images, 265 validation images, and 265 test images.

MS-COCO [8]. We additionally evaluate on images from the MS-COCO 2017 *test2017* split (41k images) to study cross-domain generalization. Since COCO does not provide annotations for the attribute taxonomies used in this work, we generate labels using Qwen3-VL-32B with a structured prompt aligned to the attribute subclasses defined by the source datasets. The hypernetwork is trained only on CLEVR-4, Stanford40, and ShotBench, and applied to COCO without further training. This setting therefore evaluates zero-shot transfer of the learned attribute transformations to a diverse real-world image distribution.

FashionIQ [16]. FashionIQ is a benchmark for composed fashion image retrieval, where each query is defined by a reference image and a natural-language change text describing how to modify it to obtain a target image. The dataset is organized into three product categories (Dresses, Shirts, Tops&Tees). We report results on the official validation split of 6,016 reference-text-target queries.

SST-5 [12]. The Stanford Sentiment Treebank (SST) is a sentence-level sentiment dataset derived from movie reviews, where each sentence is annotated with one of five ordered sentiment classes ranging from very negative to very positive. We use the standard SST-5 splits of 8,544 training and 2,210 test sentences, extracting text embeddings for each sentence and learning sentiment-conditioned transformations in the same manner as our visual-attribute experiments.

B Implementation Details

Memory Footprint Estimation. Table S6 includes a memory footprint estimation for maintaining embeddings across 14 distinct attributes. The raw storage cost for 1M ($N = 10^6$) vectors, each of dimensionality $D = 768$ and stored with 32-bit floating-point precision (4 bytes), is $N \times D \times 4$ bytes = $1,000,000 \times 768 \times 4 \approx 3.07 \approx 3$ GB. Additionally, we assume an HNSW (Hierarchical Navigable Small World) [10] index for fast search is used, incurring a memory overhead estimated at a factor of $2\times$ over the raw vector storage. This results in a total estimated in-memory footprint of $3 \text{ GB} \times 1.5 \approx 6 \text{ GB}$ for each 1M embeddings.

C Additional Experiments

Composed Image Retrieval (CIR) Compatibility. To demonstrate that our query-only transformations can *enhance* existing CIR pipelines, we augment SEARLE-XL [1] on FashionIQ [16]. We keep the gallery fixed by precomputing normalized image embeddings $\mathbf{v}_j = f_v(I_j)$ for all candidates. Given a CIR query (reference image I_r and relative caption T), SEARLE-XL produces a composed query embedding $\mathbf{q} \in \mathbb{R}^D$ in the same CLIP space. Baseline retrieval ranks images by $s_{\text{base}}(j) = \mathbf{q}^\top \mathbf{v}_j$. We then apply our *query-only* transformation using the color attribute text to obtain $\mathbf{q}'$, and define a color-aware auxiliary score $s_{\text{color}}(j) = \mathbf{q}'^\top \mathbf{v}_j$. A lightweight intent parser (Llama 3.2 3B-Instruct) maps the caption T to a signed weight $w \in [-1, 1]$ indicating whether color should be preserved ($w > 0$) or changed ($w < 0$). Final ranking uses

$$s(j) = \alpha * s_{\text{base}}(j) + (1 - \alpha) * w * s_{\text{color}}(j), \quad (1)$$

with $\alpha = 0.8$ and no training on FashionIQ. As shown in Table S1, this simple plug-in reranking consistently improves SEARLE-XL across all categories, supporting that our attribute-consistency metric is compatible with CIR and enhances performance by injecting an explicit attribute prior.

Table S1: Composed image retrieval results on FashionIQ. +**Ours** denotes augmenting SEARLE-XL [1] with our attribute-consistency metric scoring for just "color", leaving the gallery index unchanged and using no FashionIQ training.

Method	Dress		Shirt		Toptee		Average	
	R@10	R@50	R@10	R@50	R@10	R@50	R@10	R@50
SEARLE-XL [1]	25.9	42.9	18.2	37.2	26.5	45.1	23.5	41.8
+ Ours	26.3	44.9	19.8	39.0	27.8	46.1	24.6	43.3

Text-based Attribute-aware Retrieval Results. Table S2 shows that our embedding-space manipulation framework is not restricted to visual attributes. When applied to text embeddings on SST-5, learning a sentiment-conditioned transformation consistently improves performance across all evaluation modes. These gains indicate that the proposed mechanism generalizes across embedding spaces independent of the modality.

Attribute Abbreviations. For readability in tables, attribute categories are abbreviated as follows:

- Clevr-4: Col = color, Cnt = count, Shp = shape, Tex = texture.
- Stanford 40: Act = action, Loc = location, Md = mood.
- ShotBench: CA = camera angle, CM = composition, LS = lens size, LI = lighting, LT = lighting type, SF = shot framing, SS = shot size.

Expanded Attribute-aware Retrieval Results. To supplement the dataset-level retrieval results in the main paper, we also report mAP for each attribute

Table S2: Sentiment categorization on SST-5 using text embeddings (E5-large-v2). Our embedding-space transformation improves performance in all three attribute-aware tasks, demonstrating that our approach generalizes beyond vision to textual attributes.

SST5 - Sentiment			
Method	Retrieval	Clustering	Zero-Shot
Baseline	30.9	41.7	29.6
Ours	44.5	51.3	54.3

Table S3: Category-wise attribute-aware retrieval mAP across multiple datasets. Category names are abbreviated (full names in Section C). Highlighted columns denote dataset averages. Our approach retrieves meaningful samples while transforming just the query embedding.

Method	Clevr-4					Stanford 40 Action				ShotBench							
	Col	Cnt	Shp	Tex	Avg.	Act	Loc	Md	Avg.	CA	CM	LS	LI	LT	SF	SS	Avg.
Base Features	23.0	13.9	79.5	19.0	33.8	49.8	40.2	41.2	43.7	34.5	21.2	32.4	18.2	22.5	24.9	22.7	25.2
LCR	18.4	11.3	19.2	14.8	15.9	6.3	24.8	28.5	19.9	22.5	17.9	29.6	15.0	11.9	16.2	19.0	18.9
CWP	67.4	14.2	73.1	20.4	43.8	72.2	44.9	32.3	49.8	24.6	19.4	29.6	17.0	15.2	21.8	22.6	21.4
ArcFace [2]†	10.9	10.3	17.5	10.7	12.3	3.7	24.0	28.8	18.8	21.7	17.6	26.3	13.6	11.9	16.0	15.6	17.5
SupCon [6]†	13.2	13.0	14.5	10.2	12.7	3.2	24.5	28.7	18.8	22.8	17.8	27.4	13.8	11.4	15.0	15.8	17.7
StableRep+ [13]†	79.6	27.4	62.2	28.6	49.5	10.2	24.7	29.2	21.3	23.4	17.6	29.6	14.8	14.2	23.7	23.1	20.9
MaPLe [5]	23.9	16.2	70.2	23.6	33.5	45.6	44.7	53.3	47.8	32.5	20.3	30.6	17.4	20.8	23.0	21.4	23.7
LoRA [4]	34.4	17.7	83.4	36.0	42.9	46.5	46.2	55.3	49.3	30.1	17.9	31.1	14.7	12.4	24.1	23.8	22.0
Ours-Query	93.4	39.3	99.4	75.9	77.0	89.3	83.9	85.7	86.3	52.7	28.0	48.4	22.8	31.8	57.8	45.8	41.0

category in Table S3. These per-attribute measurements reveal how retrieval performance varies across attributes and demonstrate that the query transformation consistently improves attribute alignment across a broad range of properties.

Additional Datasets: Attribute-aware Retrieval. We report results on three additional benchmarks (Card, FER, and FruitSALAD) to demonstrate broad attribute coverage in Table S4. Across both query-only mode and full space inference, our method maintains strong mAP performance.

Additional Retrieval Metrics. To further validate retrieval behavior beyond mean Average Precision, we report Recall@1 and Recall@5 in Table S5. These metrics highlight how early each method surfaces an attribute-consistent result when retrieving from a frozen gallery. Across all datasets, our text-conditioned transformation maintains strong early-rank performance, reliably retrieving an attribute-matching image within the first few items.

Modified Gallery Attribute-aware Retrieval. In this setting, we remove the frozen-gallery constraint and allow each method to transform both the query and the gallery embeddings. Although this setting is less practical than the query mode, it provides a direct comparison to approaches that rely on full-space adaptation. As shown in Table S6, our learned transform, despite being trained for a query transformation setting, remains competitive with methods that train separate attribute-specific adapters. Importantly, these adapter-based approaches require one model per attribute and a corresponding recomputation

Table S4: Mean average precision (mAP) retrieval performance across other benchmarks. Top section is the query (frozen gallery) setting, bottom section is the global transformation setting. Scores are averaged across taxonomies within each dataset. Our method strongly improves retrieval across the board.

Method	Cards		FER	Fruits		All
	Rank	Suit	Emotion	Fruit	Style	Avg.
Base Features	37.9	44.6	25.6	67.7	58.8	43.4
LCR	10.4	31.0	18.1	28.1	23.1	21.5
CWP	55.5	48.0	18.5	97.6	53.3	48.6
Ours-Query	86.0	96.2	52.7	99.8	99.7	81.2
Base Features	37.9	44.6	25.6	67.7	58.8	43.4
LCR	54.6	85.4	18.7	96.8	83.4	59.6
CWP	80.2	84.9	19.1	99.1	84.5	64.5
Ours-Global	90.9	97.9	56.0	99.8	99.8	83.4

Table S5: Recall@1 and Recall@5 attribute-aware retrieval performance using a frozen gallery. Results are averaged over all categories within each dataset. Our method consistently surfaces an attribute-matching image within the top retrieved results.

Method	Clevr4		Stanford40		ShotBench	
	R@1	R@5	R@1	R@5	R@1	R@5
Base Features	70.1	88.6	72.7	92.4	40.8	77.2
LCR	19.2	34.6	9.8	48.4	17.5	47.5
CWP	51.0	69.7	61.1	81.7	21.2	50.9
ArcFace [2]†	7.9	29.4	20.2	45.0	19.3	57.6
SupCon [6]†	11.4	43.1	16.7	42.9	14.5	56.9
StableRep+ [13]†	62.0	89.1	22.1	55.8	25.3	67.0
MaPLe [5]	64.7	85.6	86.7	96.8	36.9	74.6
LoRA [4]	72.3	90.6	87.6	96.8	28.9	64.0
Ours-Query	83.8	93.5	98.7	99.8	51.0	78.1

of gallery embeddings, leading to substantially higher complexity and memory that scale linearly with the number of attributes.

Expanded Multiple-clustering Results. To complement the averaged clustering results reported in the main paper, we provide per-attribute accuracies for all datasets in Table S7. These finer-grained scores reveal how performance varies across individual attributes, offering a clear view of attribute difficulty and the consistency of the learned transformation. Across nearly all attributes, the global transform maintains stable clustering performance.

Additional Datasets: Multiple-clustering. We additionally evaluate clustering on Card, FER, and FruitSALAD to demonstrate broad attribute coverage in Table S8. In all three benchmarks, the global transformation produces stable and competitive clustering accuracy, confirming that the same attribute-aware transform generalizes effectively across diverse visual domains.

Table S6: Mean average precision (mAP) retrieval performance across all benchmarks allowing for transformed galleries. Scores are averaged across taxonomies within each dataset. Our method remains competitive with attribute-specific trained models. Memory footprint details explained in Section B.

Method	Memory	Clevr4	Stanford40	ShotBench
	Footprint	mAP	mAP	mAP
Base Features	6GB	33.8	43.7	25.2
LCR	84GB	52.8	46.7	22.0
CWP	84GB	55.6	53.2	23.2
ArcFace [2]†	84GB	86.4	61.1	38.8
SupCon [6]†	84GB	89.1	66.3	42.6
StableRep+ [13]†	84GB	88.8	60.8	43.1
MaPLe [5]	84GB	50.4	48.6	23.9
LoRA [4]	84GB	79.9	54.8	25.1
Ours-Global	6GB	85.6	95.7	41.9

Table S7: Category-wise clustering accuracy results across multiple datasets. Category names are abbreviated (full names in Section C). Highlighted columns denote averages across taxonomies. Our approach achieves strong, consistent performance across both semantic and perceptual visual attributes.

Method	Clevr-4					Stanford 40 Action				ShotBench							
	Col	Cnt	Shp	Tex	Avg.	Act	Loc	Md	Avg.	CA	CM	LS	LI	LT	SF	SS	Avg.
Base Features	13.1	13.1	89.2	13.9	32.3	59.4	58.6	75.8	64.6	51.9	28.2	38.5	23.3	26.8	30.3	28.4	32.5
LCR	85.9	28.3	81.9	29.4	56.4	64.8	65.5	64.2	64.8	35.8	24.2	37.5	25.8	25.5	27.8	34.8	30.2
CWP	80.2	35.6	97.3	28.0	60.3	74.8	59.1	60.8	64.9	37.2	27.1	38.3	25.8	29.8	33.9	33.4	32.2
Multi-Map _{CVPR'24}	75.3	53.9	65.5	56.5	62.8	57.9	59.4	71.0	62.8	45.1	28.0	38.2	23.8	31.0	29.9	31.9	32.6
Multi-Sub _{NeurIPS'25}	84.7	63.3	74.9	65.9	72.2	60.4	63.1	74.7	66.1	44.0	28.0	37.9	24.0	29.9	31.4	31.3	32.4
IC TC CLR ₂₄	53.4	43.5	71.9	62.8	57.9	77.7	75.0	75.5	76.1	27.5	18.0	31.6	20.3	13.7	18.0	23.1	21.7
SSD-LLM _{ECCV'24}	49.1	44.2	72.0	60.8	56.5	81.4	70.4	70.6	74.1	26.7	23.5	29.1	23.2	20.1	29.0	23.4	25.0
X-Cluster _{arxiv'25}	70.3	65.7	58.4	65.3	64.9	82.8	69.8	52.3	68.3	-	-	-	-	-	-	-	-
Ours-Global	72.1	59.0	87.6	73.5	73.0	84.4	67.3	88.3	80.0	65.9	32.6	61.9	25.9	38.9	74.6	61.8	51.6

Conditional Similarity Networks Comparison. Conditional Similarity Networks (CSN) [15] trains attribute masks while tuning ResNet-18, not satisfying our frozen-gallery constraint. We compare against CSN’s published performance on Zappos50k [15] and against an implementation of their method with a frozen-CLIP-backbone. Table S9 shows our method wins each comparison, with the widest margins on the more semantic Stanford-40 dataset.

Hyperparameter Sensitivity Analysis. We evaluate the effect of the multi-modal guidance weight ω on retrieval and clustering performance in Table S10. Moderate guidance improves retrieval while maintaining similar clustering performance, with $\omega = 0.1$ performing best overall and therefore used in our experiments. As ω increases further, clustering accuracy improves, while retrieval performance begins to degrade. This behavior suggests that stronger text alignment increasingly reshapes the embedding geometry, which can benefit clustering structure but reduce the fine-grained similarity relationships needed for retrieval.

Table S8: Multiple clustering performance across more benchmarks, averaged across taxonomies within each dataset. Our method strongly improves clustering accuracy.

Method	Cards		FER	Fruits		All
	Rank	Suit	Emotion	Fruit	Style	Avg.
Base Features	30.4	63.1	27.0	52.7	33.1	38.9
LCR	56.5	93.5	23.0	86.4	86.5	61.5
CWP	71.9	92.3	23.9	71.3	80.8	60.7
Ours-Global	78.5	99.2	65.7	85.3	100.0	82.4

Table S9: Comparison to Conditional Similarity Networks adapted baseline. Our method wins each comparison.

Dataset	CSN (reported) w/ ResNet-18	CSN (our impl.) w/ CLIP	Ours
Clevr-4 (mAP)	—	70.6	77.6
Stanford-40 (mAP)	—	46.5	84.4
Zappos (triplet acc.)	78.8	80.9	85.4

Does the transformation preserve general visual structure beyond the target attribute?

Although the transformed embeddings primarily emphasize the desired attribute, we find that meaningful global relationships remain intact. Table S11 shows that our transformation does not collapse or distort the underlying feature space merely to satisfy the contrastive loss. Text-based classification for unseen class names drops only slightly, and k-NN evaluations confirm that relative similarity is preserved—samples that are visually and semantically similar remain close. For example, a “blue airplane” remains closer to other blue airplanes than to blue cars, although it is closer to both compared to a “yellow airplane”. This indicates that the transformation sharpens attribute-specific structure without erasing general semantics.

Cross-dataset generalization on MS-COCO. To further evaluate generalization, we test the trained model on images from the MS-COCO test2017 split without any additional training. Since COCO does not contain the attribute taxonomies used in our experiments, we automatically annotate images with the relevant attributes and subclasses using Qwen3-VL-32B with a structured prompting scheme. The trained hypernetwork remains unchanged and is applied directly to the COCO embeddings.

Table S12 reports attribute-aware retrieval performance under this setting. Despite the relatively small size of the training datasets and a notable distri-

Table S11: Zero-shot classification and k-NN evaluation (k=5) on CIFAR10. The original class names are not captured in unrelated “color” transformation, yet k-NN evaluation shows that *relative* similarities are preserved.

Method	CIFAR10	
	ZS Acc.	k-NN
Base Features	86.06	97.66
Ours + “color”	52.57	97.48

Table S10: Effect of multimodal guidance weight ω on CLEVR-4 retrieval (mAP) and clustering accuracy. Moderate guidance provides the best overall performance, while larger values slightly improve clustering at the expense of retrieval.

	Loss	Weight	Retrieval	Clustering
$\omega = 0$			72.3	78.8
$\omega = 0.01$			72.4	79.6
$\omega = 0.1$			73.4	79.6
$\omega = 1$			73.0	81.5
$\omega = 10$			69.9	81.6

bution shift between the training datasets and COCO (e.g., Clevr-4 provides synthetic object images for learning color and count attributes), our method consistently improves over the baseline across all attribute categories. These results suggest that the learned transformations capture transferable attribute structure encoded in the foundation model embeddings rather than overfitting to the specific training datasets.

Table S12: Zero-shot attribute retrieval performance on MS-COCO. A model trained only on the source datasets (CLEVR-4, Stanford40, and ShotBench) is evaluated on COCO images annotated with attribute labels for evaluation. Results show consistent improvements across all attributes, indicating that the learned transformations capture transferable attribute structure beyond the training distributions.

Method	Action	Location	Mood	Color	Count	Camera	Angle	Shot	Size
Baseline	21.9	36.6	34.9	19.7	30.5	36.4		30.7	
Ours - Query	27.4	47.1	37.4	24.7	34.8	42.0		39.3	

Qualitative Results. Beyond quantitative evaluations, we visualize the behavior of our text-conditioned transformations across a range of attributes and visual domains. Figure S1 visualizes ranked retrieval results on the Fruit-SALAD dataset using the “style” attribute. Our transform causes the retrieval to respect the input attribute first, retrieving different fruit images in the same style before retrieving same fruit images in a different style. Figure S2 highlights retrievals driven by the shot framing attribute, where our method prioritizes compositional structure over object identity. Figure S3 demonstrates robustness to stylistic variation: even with a stylized drawing as input, the “emotion” transform retrieves semantically aligned expressions. Finally, Figure S4 illustrates emergent compositional control by showing that color, location, and their combination produce distinct and semantically coherent retrieval sets. Collectively, these examples show that the learned transformations cleanly isolate visual attributes, generalize across domains, and support multi-attribute control.

Goal: retrieve images with similar art styles

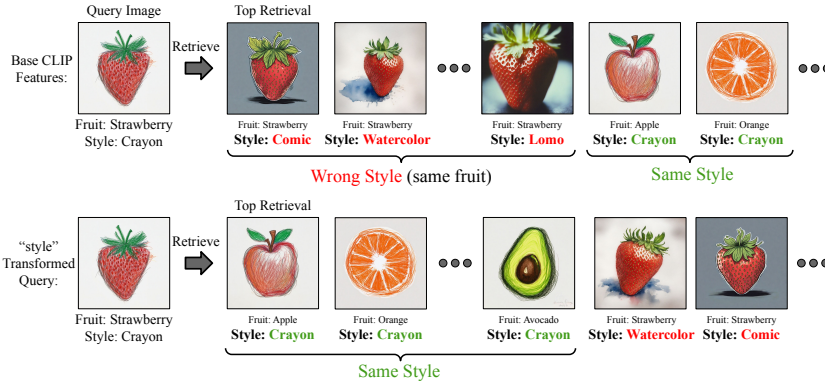


Fig. S1: Retrieval ranked by similarity using base CLIP embeddings and our transformed query (“style”). CLIP prioritizes object semantics (e.g., fruit type), whereas our transformation maintains attribute alignment—retrieving different fruits rendered in the same artistic style.

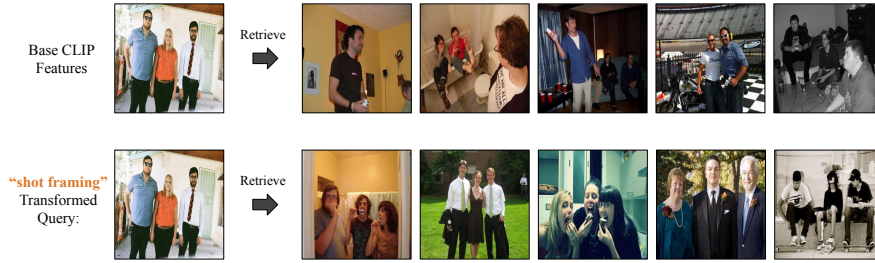


Fig. S2: Top retrieval results comparing CLIP embeddings and our text-guided transformation using the query “shot framing.” Our method surfaces images with matching three-subject shot-framing structure.



Fig. S3: Attribute-aware retrieval using the “emotion” query. Despite the input being a stylized drawing, the learned transform isolates the intended emotion and retrieves aligned “angry” expressions across diverse visual styles.

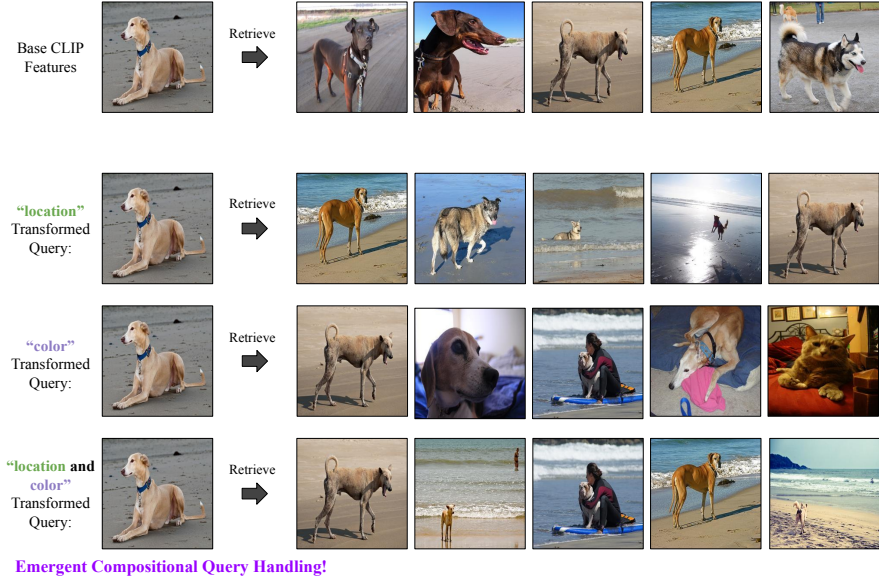


Fig. S4: Retrieval comparison illustrating compositional attribute control. CLIP primarily retrieves dog-related images, whereas our transformations isolate specific attributes: the “color” transform finds dogs of similar color, the “location” transform retrieves dogs on beaches, and the combined query “location and color” retrieves beach scenes with dogs of similar coloration, demonstrating emergent multi-attribute controllability.

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